



HUMAN EYE -05

Class 10 - Science

Section A

1. **Assertion (A):** A person suffering from myopia cannot see the distant objects clearly. [1]
Reason (R): A converging lens is used for the correction of myopic eye as it can form real as well as virtual images of the objects placed in front of it.

a) Both A and R are true and R is the correct explanation of A.	b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.	d) A is false but R is true.
2. **Assertion (A):** The image of a distant object is formed in front of the retina and not at the retina itself. [1]
Reason (R): The excessive curvature of the eye lens is a defect of myopia.

a) Both A and R are true and R is the correct explanation of A.	b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.	d) A is false but R is true.
3. **Assertion (A):** It is possible to correct the refractive defects with contact lenses or through surgical interventions. [1]
Reason (R): The eye lens is composed of fibrous jelly-like material.

a) Both A and R are true and R is the correct explanation of A.	b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.	d) A is false but R is true.
4. **Assertion (A):** Cataract causes partial or complete loss of vision. [1]
Reason (R): The eyeball is approximately elliptical shape.

a) Both A and R are true and R is the correct explanation of A.	b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.	d) A is false but R is true.
5. **Assertion (A):** Sometimes, the eye may gradually lose its power of accommodation. [1]
Reason (R): The crystalline lens of people at old age becomes milky and cloudy.

a) Both A and R are true and R is the correct explanation of A.	b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.	d) A is false but R is true.
6. **Assertion (A):** There exist two angles of incidence for the same magnitude of deviation (except minimum [1]

deviation) by a prism kept in the air.

Reason (R): In a prism kept in the air, a ray is incident on the first surface and emerges out of the second surface. Now if another ray is incident on the second surface (of the prism) along the previous emergent ray, then this ray emerges out of the first surface along the previous incident ray. The principle is called the principle of reversibility of light.

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|---|---|
| a) Both A and R are true and R is the correct explanation of A. | b) Both A and R are true but R is not the correct explanation of A. |
| c) A is true but R is false. | d) A is false but R is true. |

7. **Assertion (A):** Thin prisms do not deviate light much. [1]

Reason (R): Thin prism have small angle A and hence, $D_m = [(\mu - 1) A]$, where μ is the refractive index of prism w.r.t. medium 1.

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|---|---|
| a) Both A and R are true and R is the correct explanation of A. | b) Both A and R are true but R is not the correct explanation of A. |
| c) A is true but R is false. | d) A is false but R is true. |

8. **Assertion (A):** Higher the refractive index of the prism material, lower is the angle of deviation. [1]

Reason (R): The angle of deviation is directly proportional to the angle of prism.

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| a) Both A and R are true and R is the correct explanation of A. | b) Both A and R are true but R is not the correct explanation of A. |
| c) A is true but R is false. | d) A is false but R is true. |

9. **Assertion (A):** Sunlight reaches us without dispersion in the form of white light and not as its components. [1]

Reason (R): Dispersion takes place due to variation of refractive index for different wavelengths but in a vacuum, the speed of light is independent of wavelength and hence vacuum is a non-dispersive medium.

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|---|---|
| a) Both A and R are true and R is the correct explanation of A. | b) Both A and R are true but R is not the correct explanation of A. |
| c) A is true but R is false. | d) A is false but R is true. |

10. **Assertion (A):** The Sun appears flattened at sunrise and sunset. [1]

Reason (R): The apparent flattening of the Sun's disc at sunrise and sunset is due to atmospheric refraction.

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|---|---|
| a) Both A and R are true and R is the correct explanation of A. | b) Both A and R are true but R is not the correct explanation of A. |
| c) A is true but R is false. | d) A is false but R is true. |

11. **Assertion (A):** The twinkling of stars is due to the fact that the refractive index of the earth's atmosphere fluctuates. [1]

Reason (R): In cold countries, the phenomenon of looming (i.e., ship appears in the sky) takes place, because the refractive index of air decreases with height.

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| a) Both A and R are true and R is the correct explanation of A. | b) Both A and R are true but R is not the correct explanation of A. |
| c) A is true but R is false. | d) A is false but R is true. |

12. **Assertion (A):** The twinkling of stars is due to the fact that the refractive index of the earth's atmosphere [1]

fluctuates.

Reason (R): When light propagates from one medium to another its direction of propagation changes.

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|---|---|
| a) Both A and R are true and R is the correct explanation of A. | b) Both A and R are true but R is not the correct explanation of A. |
| c) A is true but R is false. | d) A is false but R is true. |

13. **Assertion (A):** The stars twinkle while the planet do not. [1]

Reason (R): The stars are much bigger in size than the planets.

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|---|---|
| a) Both A and R are true and R is the correct explanation of A. | b) Both A and R are true but R is not the correct explanation of A. |
| c) A is true but R is false. | d) A is false but R is true. |

14. **Assertion (A):** The blue colour of the sky appears due to the scattering of blue colour. [1]

Reason (R): Blue colour has the shortest wavelength in the visible spectrum.

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| a) Both A and R are true and R is the correct explanation of A. | b) Both A and R are true but R is not the correct explanation of A. |
| c) A is true but R is false. | d) A is false but R is true. |

15. **Assertion (A):** Danger signals are made of red colour. [1]

Reason (R): Wavelength of red is maximum and hence scattering of red colour is minimum.

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| a) Both A and R are true and R is the correct explanation of A. | b) Both A and R are true but R is not the correct explanation of A. |
| c) A is true but R is false. | d) A is false but R is true. |

16. **Assertion (A):** Danger signals are red. [1]

Reason (R): Red colour has smallest wavelength.

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|---|---|
| a) Both A and R are true and R is the correct explanation of A. | b) Both A and R are true but R is not the correct explanation of A. |
| c) A is true but R is false. | d) A is false but R is true. |

17. **Assertion (A):** The scattered light makes path of light visible. [1]

Reason (R): Scattering of light is the result of Tyndall effect.

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| a) Both A and R are true and R is the correct explanation of A. | b) Both A and R are true but R is not the correct explanation of A. |
| c) A is true but R is false. | d) A is false but R is true. |

18. **Assertion (A):** Danger signals are made of red colour. [1]

Reason (R): Velocity of red light in air is maximum, so signals are visible even in dark.

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| a) Both A and R are true and R is the correct explanation of A. | b) Both A and R are true but R is not the correct explanation of A. |
| c) A is true but R is false. | d) A is false but R is true. |

19. **Assertion (A):** The Sun appears red during sunrise or sunset. [1]

Reason (R): The scattering of light is inversely proportional to its wavelength.

- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
20. **Assertion (A):** The sky appears dark to people flying at high altitudes. [1]
Reason (R): The atmosphere is denser close to the earth.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
21. **Assertion (A):** The sky looks dark and black instead of blue in outer space. [1]
Reason (R): There is no air in the outer space to scatter the sunlight.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
22. **Assertion (A):** The scattering of longer wavelengths of light increases as the size of the particles increases. [1]
Reason (R): Large particles scatter lights of all wavelengths equally well.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
23. **Assertion (A):** The phenomenon of scattering of light by the colloidal particles gives rise to the Tyndall effect. [1]
Reason (R): The colour of the scattered light depends on the size of the scattering particles.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
24. **Assertion (A):** The blue colour of the sky appears due to the scattering of blue colour. [1]
Reason (R): Blue light has a longer wavelength.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
25. **Assertion (A):** The focal length of the mirror is f and the distance of the object from the focus is u . The magnification of the mirror is $\frac{f}{u}$. [1]
Reason (R): Magnification = $+\frac{\text{Size of image}}{\text{Size of object}}$
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
26. **Assertion (A):** The resolving power of a telescope is more if the diameter of the objective lens is more. [1]
Reason (R): Objective lens of large diameter coiled more light.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.

explanation of A.

correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

Section B

27. Which of the following statements is correct regarding the propagation of light of different colours of white light in air? [1]

a) All the colours of the white light move with the same speed

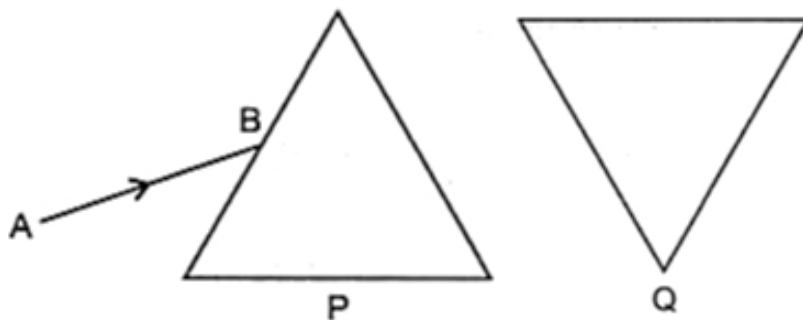
b) Red light moves fastest

c) Blue light moves faster than green light

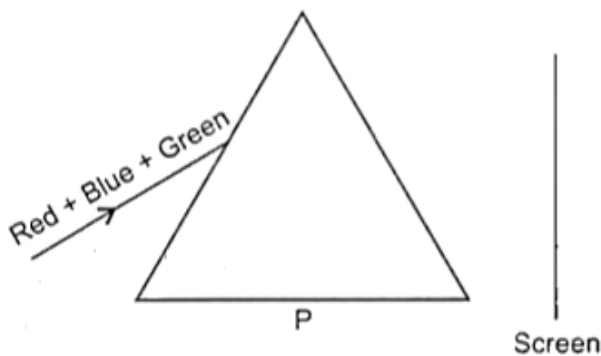
d) Yellow light moves with the mean speed as that of the red and the violet light

Section C

28. The given figure shows two identical prisms P and Q placed with their faces parallel to each other. A light ray of yellow colour AB is incident at the face of the prism P. Complete the diagram to show the path of the ray till it emerges out of the prism Q. [2]



29. The figure shows a light (red + blue + green) incident on a prism P. Complete the diagram by drawing the refracted and the emergent rays. [2]



30. Differentiate between a glass slab and a glass prism. What happens when a narrow beam of [2]

i. a monochromatic light, and

ii. white light passes through

a. glass slab and

b. glass prism?

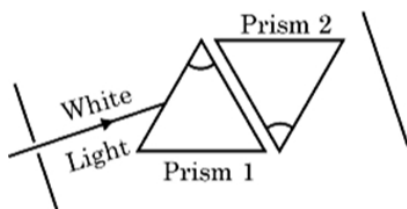
31. i. With the help of labelled ray diagram show the path followed by a narrow beam of monochromatic light when it passes through a glass prism. [2]

ii. What would happen if this beam is replaced by a narrow beam of white light?

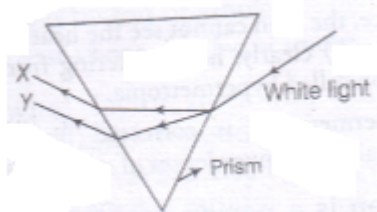
32. i. State the relation between colour of scattered light and size of the scattering particle. [2]

ii. The apparent position of an object, when seen through the hot air, fluctuates or wavers. State the basic cause of this observation.

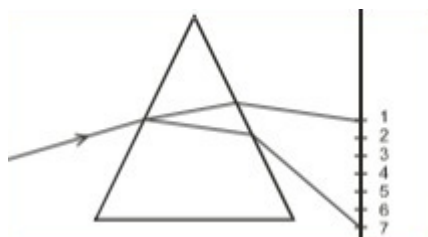
iii. Complete the path of white light when it passes through two identical prisms placed as shown:



33. Draw a neat diagram to show refraction of a light ray through a triangular glass prism. Mark angle of incidence, angle of emergence, incident ray, refracted ray, emergent ray and the angle of deviation. [2]
34. When a beam of white light is passed through a triangular glass prism, it gets dispersed into its seven colour components. Why do we get these colours? In the given figure, the colours X and Y represent the extreme components of the spectrum. Identify X and Y. [2]



35. Draw a ray diagram showing the dispersion through a prism when a narrow beam of white light is incident on one of its refracting surfaces. Also indicate the order of the colour of the spectrum obtained. [2]
36. Why do different colours get separated when a white light passes through a prism? How can we recombine the components of white light after a prism has separated them? Explain with the help of figure. [2]
37. Why we see a rainbow just after rains? [2]
38. When and where do we see a rainbow? How is a rainbow formed? Draw a labelled diagram to illustrate the formation of a rainbow. [2]
39. Explain the phenomenon of dispersion of white light through a glass prism, using suitable ray diagram. [2]
40. A beam of white light falling on a glass prism gets split up into seven colours marked 1 to 7 as shown in the diagram. [2]

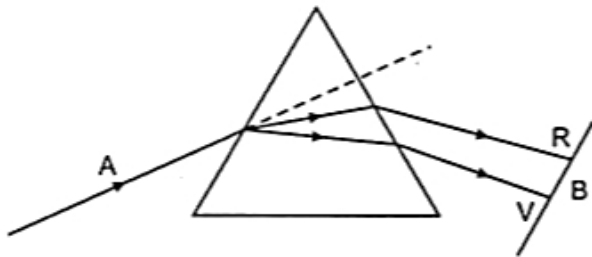


- i. The colour at position marked 3 and 5 are similar to the colour of the sky and the colour of gold metal, respectively. Is the above statement made by the student correct or incorrect justify? [2]
- ii. Which of the above shown positions corresponds approximately to the colour of [2]
- a brinjal,
 - danger signal,
 - neel which is applied to clothes,
 - orange.
41. Why do the component colours of incident white light split into a spectrum while passing through a glass prism, explain. [2]
42. What is dispersion? What happens when light is passed through a glass prism? [2]
43. a. A narrow beam of white light is incident on three glass objects as shown below. Comment on the nature of behaviour of the emergent beam in all three cases. [2]



b. There is a similarity between two of the emergent beam. Identify the two.

44. How will you use two identical prisms so that a narrow beam of white light incident on one prism emerges out of the second prism as white light? Draw the diagram. [2]
45. Draw a labelled diagram to show (i) dispersion of a beam of white light, and (ii) formation of a rainbow. [2]
46. What happens to a beam of white light when it gets refracted through glass prism? Which colour deviates the most and the least after refraction through a prism? What is likely to happen if a second identical prism is placed in an inverted position with respect to the first prism. Justify your answer. [2]
47. Why do we see a rainbow in the sky only after rainfall? [2]
48. Name four colours of the spectrum of white light which have wavelengths longer than blue light. [2]
49. What phenomenon is depicted in the given diagram? Explain the phenomenon and label A and B in the diagram. [2]



50. Define the term dispersion of white light. State the colour which bends (i) the most, (ii) the least while passing through a glass prism. Draw a diagram to show the dispersion of white light. [2]
51. What is meant by the dispersion of white light? Draw a diagram to show dispersion of white light by the glass prism. [2]
52. Why do clouds appear white? Explain. [2]
53. Name the phenomenon responsible for [2]
 - i. advance sunrise
 - ii. colour of water in deep sea
 - iii. twinkling of stars
 - iv. reddish appearance of the Sun at the time of sunrise and sunset.
54. Explain why the planets do not twinkle. [2]
55. Why do stars twinkle? [2]
56. Is the position of a star as seen by us its true position? Justify your answer. [2]
57. Give reasons: [2]
 - i. Red colour is selected for danger signals.
 - ii. The sky appears dark in space.
 - iii. The time difference between the actual sunset and apparent sunset is about 2 minutes.
58. When sunlight passes through the canopy of a dense forest, the light is scattered. Explain. [2]
59. Why does the sky appears dark instead of blue to an astronaut? [2]
60. Why does the sky appear dark instead of blue to astronaut? [2]
61. Using the following informations form a pathway to explain the phenomenon behind the colour of the sun at sunrise and sunset. And also include informations that are not mentioned below to complete it. White light, blue light, shorter wavelength, larger wavelength, and horizon. [2]
62. Explain, why the Sun overhead at noon, appears white? [2]

63. Under very dim light, we are able to see all the objects but cannot distinguish between their colours. Explain. [2]
64. Give reasons for the following : [2]
- Danger signals installed at airports and at the top of tall buildings are of red colour.
 - The sky appears dark to the passengers flying at very high altitudes.
 - The path of a beam of light passing through a colloidal solution is visible.
65. Why does the sky at noon appear white? [2]
66. A reporter records the following observations of an astronaut from his space ship. [2]
- The length of the day is same as observed on the earth.
 - Sky appears black in colour.
 - The star appears to twinkle while the planets do not do so as they do on the earth.
- Justify each statement
67. I. Draw a figure which show the arrangement for observing the phenomenon of scattering of light in the laboratory. [2]
- II. What colours would you observe in the experiment? Why?
68. Explain, how the scattering of light depends upon particle size? [2]
69. What is the difference in colours of the Sun observed during sunrise/sunset and noon? Give explanation for each. [2]
70. Why is it difficult to drive on a foggy day? [2]
71. What is the colour of an emergent light when white light is passed through a colloidal solution of sulphur? [2]
- Explain.
72. What is scattering of light? Explain how the colour of the scattered light depends on the size of the scattering particles. [2]
73. In today's digital world, every human spends around a minimum eight hours with the digital electronic items which make their eyes to strain more and causes many eye problems. Still they undergo many exercises for their eyes during their working hours what would be your medical suggestion to reduce the stress and strain caused in the eyes? [2]